MINERvA OmniFold

Research discussion with Ben

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2D validation, higher-dimensional inference, and full-event learning

11 October 2026

Reproduction, conditional inference and PET development

2D central result	Published muon cross section reproduced. XR now reproduces both LightGBM reference centrals.
Joint 5D inference	Ten specified generator hypotheses rejected under a calibrated hybrid model. Measurement branch did not qualify.
Full-event PET	Recovery improves for some truth changes. Tested intervals still fail in low-acceptance regions.

Reproduction, model rejection and a calibrated measurement are different achievements.

XR interim snapshot: 11 October, 11:20 UTC. Covariance transfer and corrected-band coverage remain open.

The selected neutrino sample and its truth domain

Sample	Medium-energy neutrino mode, about 6 GeV. 4.12 million selected events
Signal	Muon-neutrino charged-current interactions in the tracker
Reference domain	Muon angle $< 20°$, $1.5 < p_{\parallel} < 60$ GeV/c, $pT < 4.5$ GeV/c
Detector	MINERvA measures activity. Downstream MINOS measures the muon

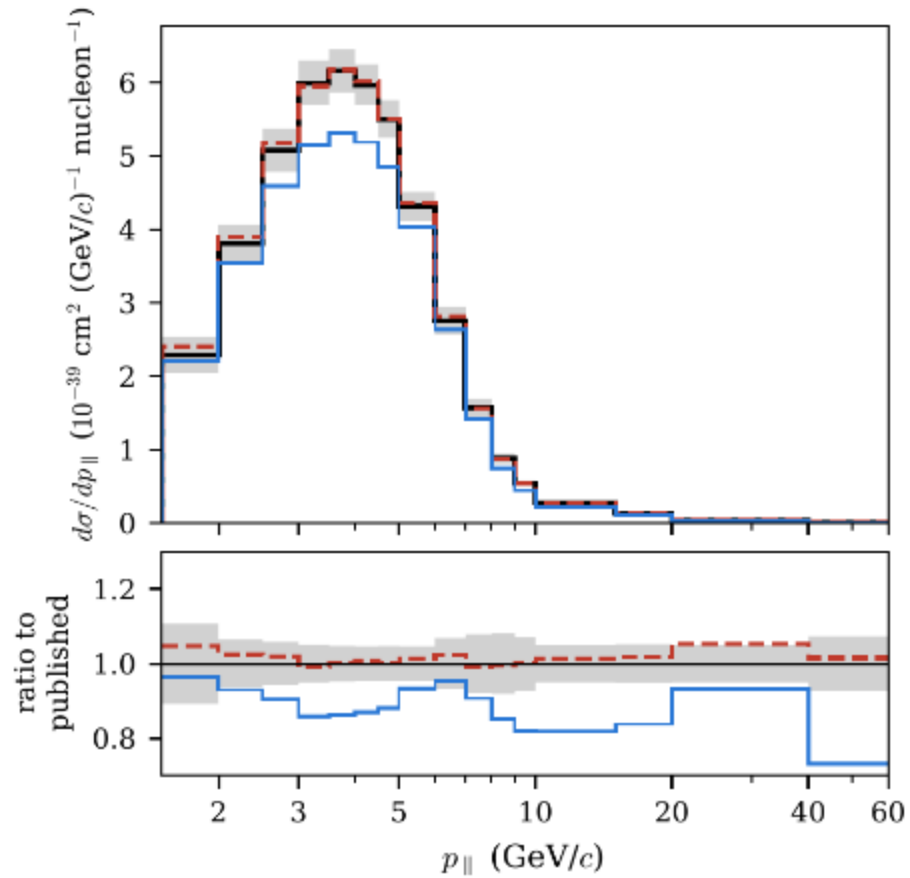
Truth-only events remain in the signal denominator even when reconstruction misses them.

Closure, dimensional anchors and coverage

Closure	Create simulated data with known truth and test recovery
Dimensional anchor	Integrate an added observable and compare with the earlier result
Coverage	Repeat simulated experiments and count how often intervals contain truth

A closure pass applies to the tested deformation and detector response

The 2D central result reproduces the published scale



Black: published Red dashed: OmniFold Blue: tuned prior

3.073 × 10^{-38} cm^2/nucleon

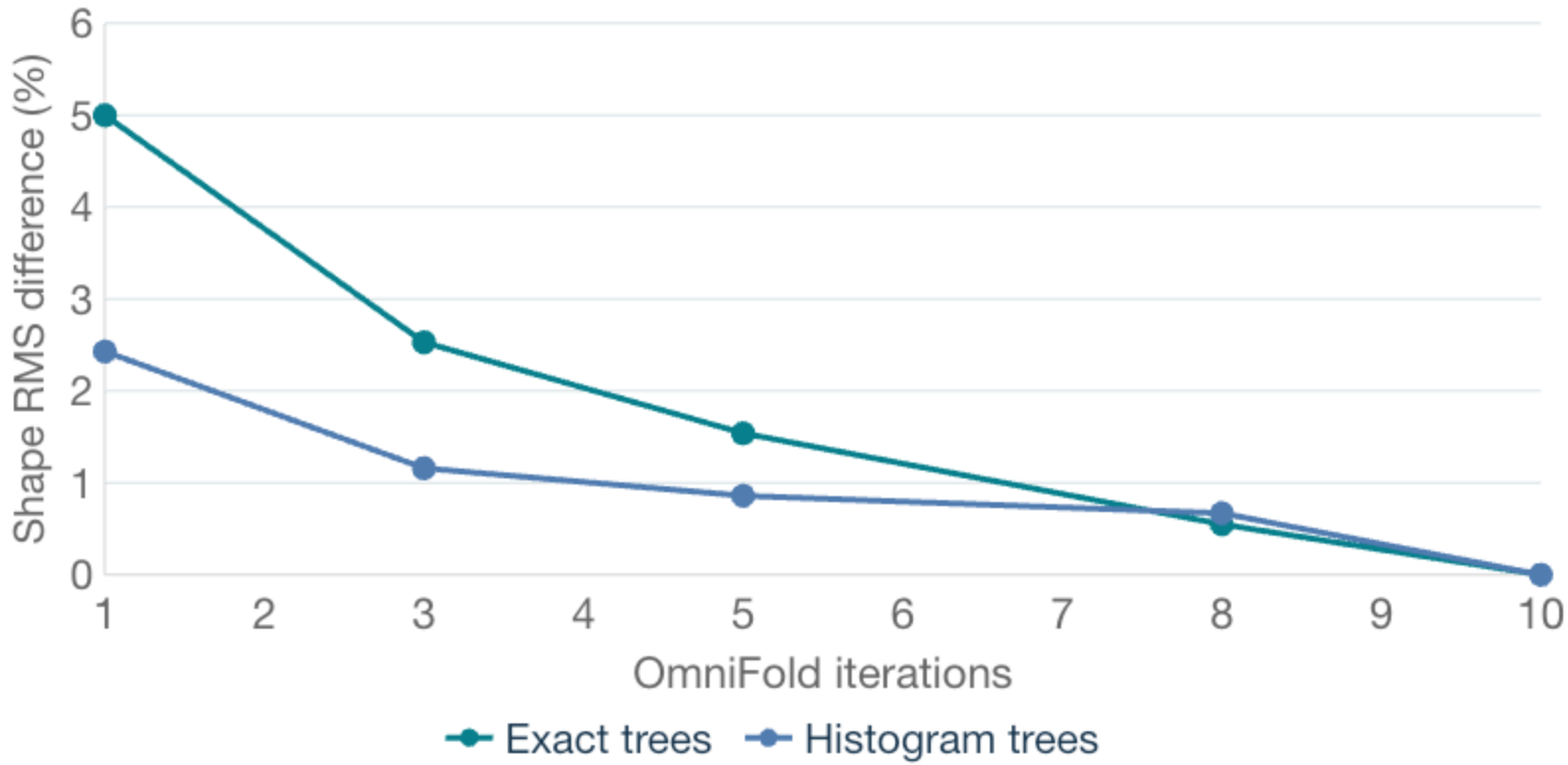
1.11% above the published total

205 reported bins. Five iterations

Same data and shared systematics

Projection of the validated 2D central result. This is a reproduction, not an independent measurement

Most iteration movement occurs before five iterations



1.54% at k = 5

Exact-tree shape difference from its k = 10 reference.

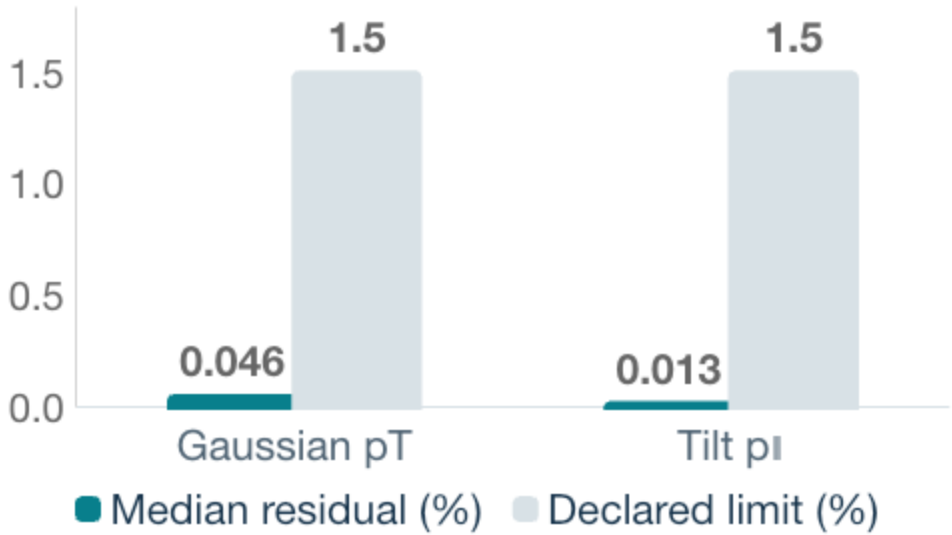
Reported total change from 5 to 10: 0.026%.

Five iterations capture most of the movement. Residual shape evolution remains.

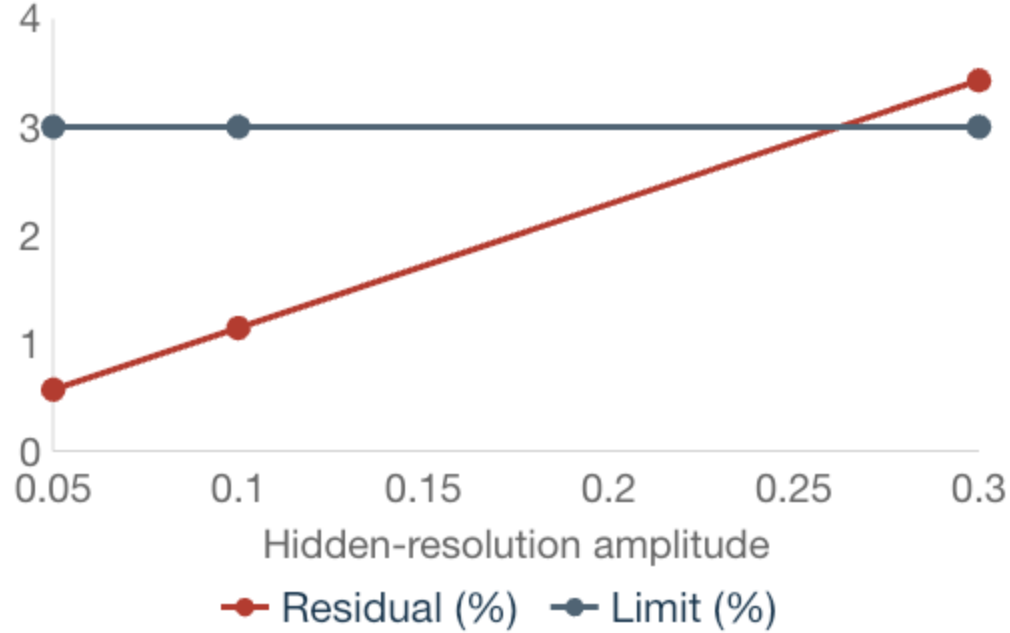
Playlist 1A. Each curve uses its own k = 10 reference, whose zero is automatic. Tabulated archived scan.

Closure passes depend on what the test changes

Changed truth shape



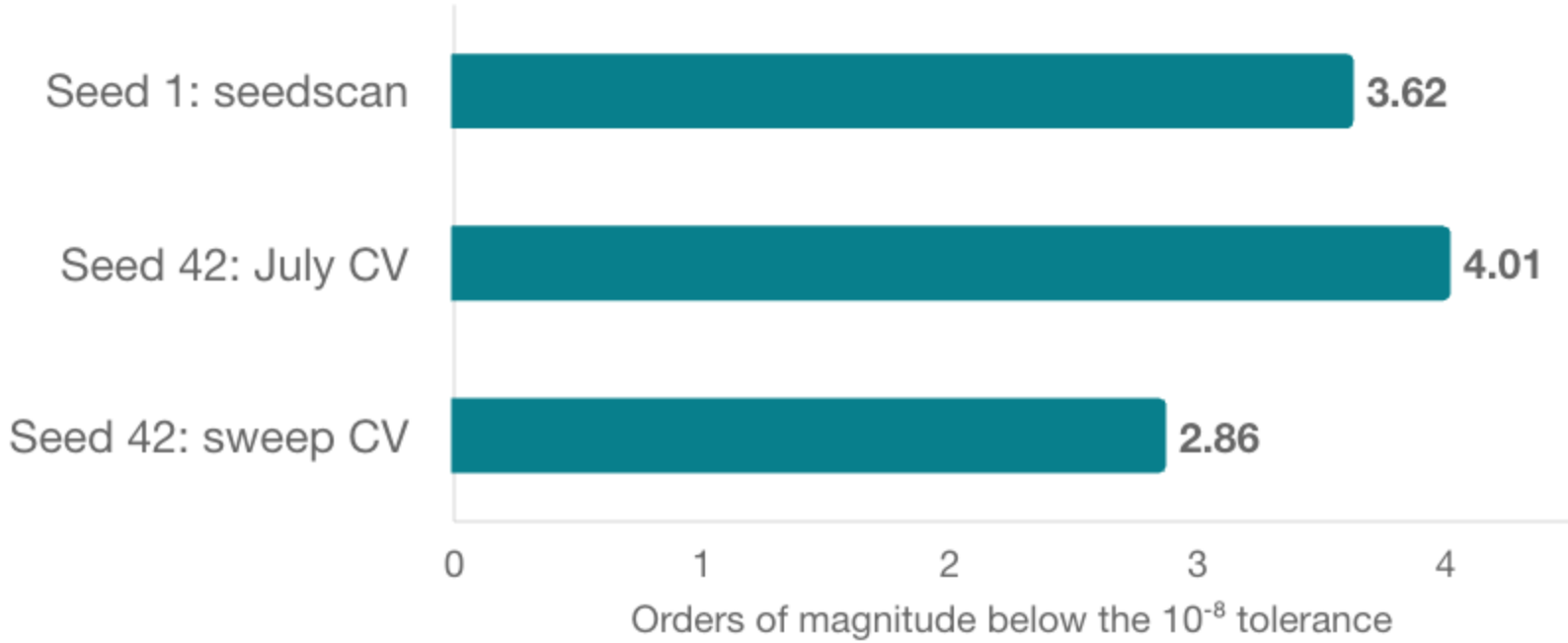
Changed hidden resolution



Easy shapes close. A hidden-resolution departure exceeds its limit at amplitude 0.30.

1A, LightGBM, 5 iterations. Hidden variable: ΔpT = reco pT – truth pT, absent from the classifier inputs.

XR reproduces both LightGBM reference centrals



All 205 cells pass

Seed 1: 2.41×10^{-12}
Seed 42 July: 9.81×10^{-13}
Seed 42 sweep: 1.38×10^{-11}

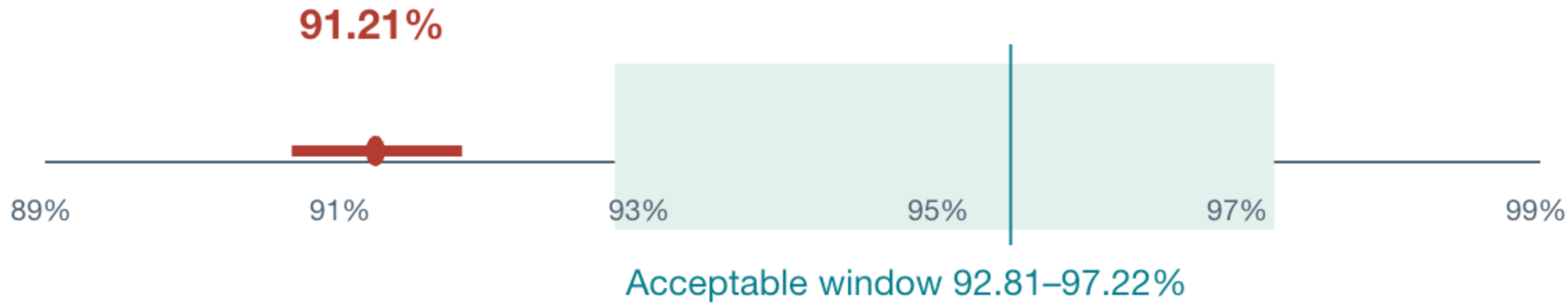
Criterion: 10^{-8}

Reference reproduction is established for LightGBM. Transfer to the exact-tree central remains unmeasured.

Committed interim XR results, 9be8bfb4. Exact-tree comparisons incomplete at 11:20 UTC; final campaign review pending.

The repaired 2D band still needs a coverage test

Old band: nominal 95.45% containment



99.5% confidence interval: 90.65–91.79%

VL170 repaired completeness. Its independent coverage re-test is still outstanding.

Old VL162 band, 100 fixed-truth toys. XR tests reproducibility. The later split-sample design establishes only conditional width agreement.

Added coordinates expose hadronic structure

Muon kinematics

pT and p∥ anchor the published result.

Visible hadronic energy

Eavail records selected visible energy, with a species-defined sum.

Calorimetric transfer

q0 supplies recoil information used to derive q3 and W.

2D **+ Eavail** **+ q3, W**

Hadronic observables reveal structure hidden by muon-only projections.

Eavail differs from q0. Outgoing neutrinos and neutrons contribute zero to the implemented Eavail sum.

Reconstructed q3 and W share the same event inputs

Measured inputs: muon energy / direction and calorimetric q0

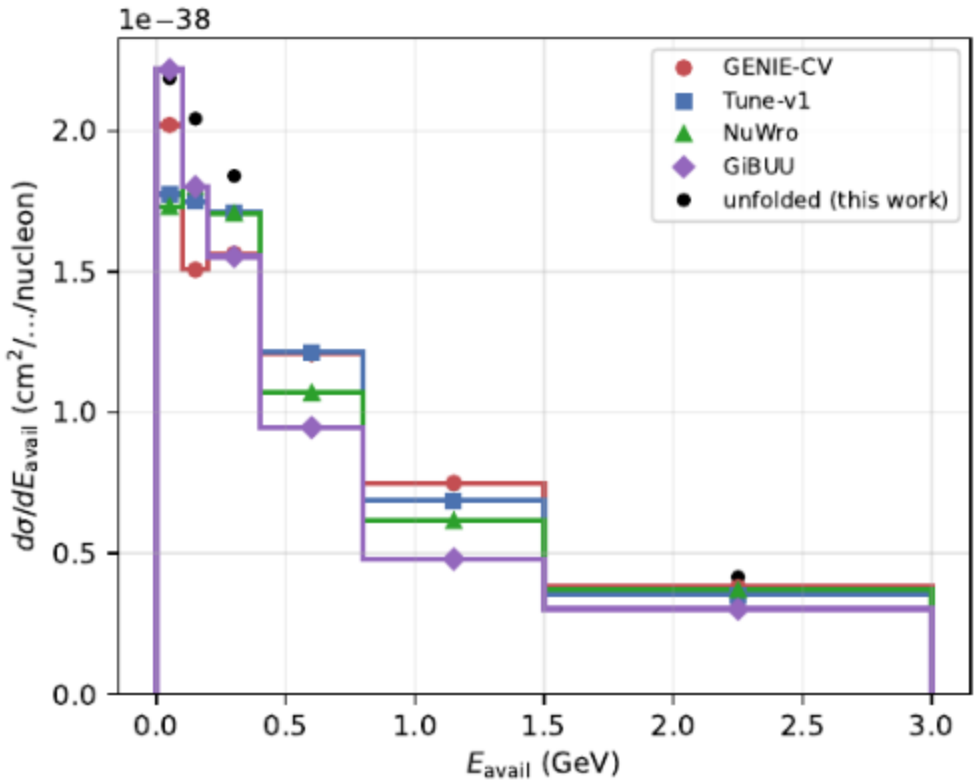
$$E_\nu = E_\mu + q0 \qquad Q^2 = 2E_\nu(E_\mu - p_\mu \cos\theta_\mu) - m_\mu^2$$

$q3^2 = Q^2 + q0^2$ **$W^2 = M^2 + 2Mq0 - Q^2$**

W adds a useful physics coordinate, but no independent reconstructed input once muon kinematics and q3 are given.

q0: full calorimetric transfer. M: average nucleon mass. Physical interior, $q0 \geq 0$. Code clips Q^2 and W^2 at zero.

Hadronic energy reveals model differences



3D normalization anchors to 2D

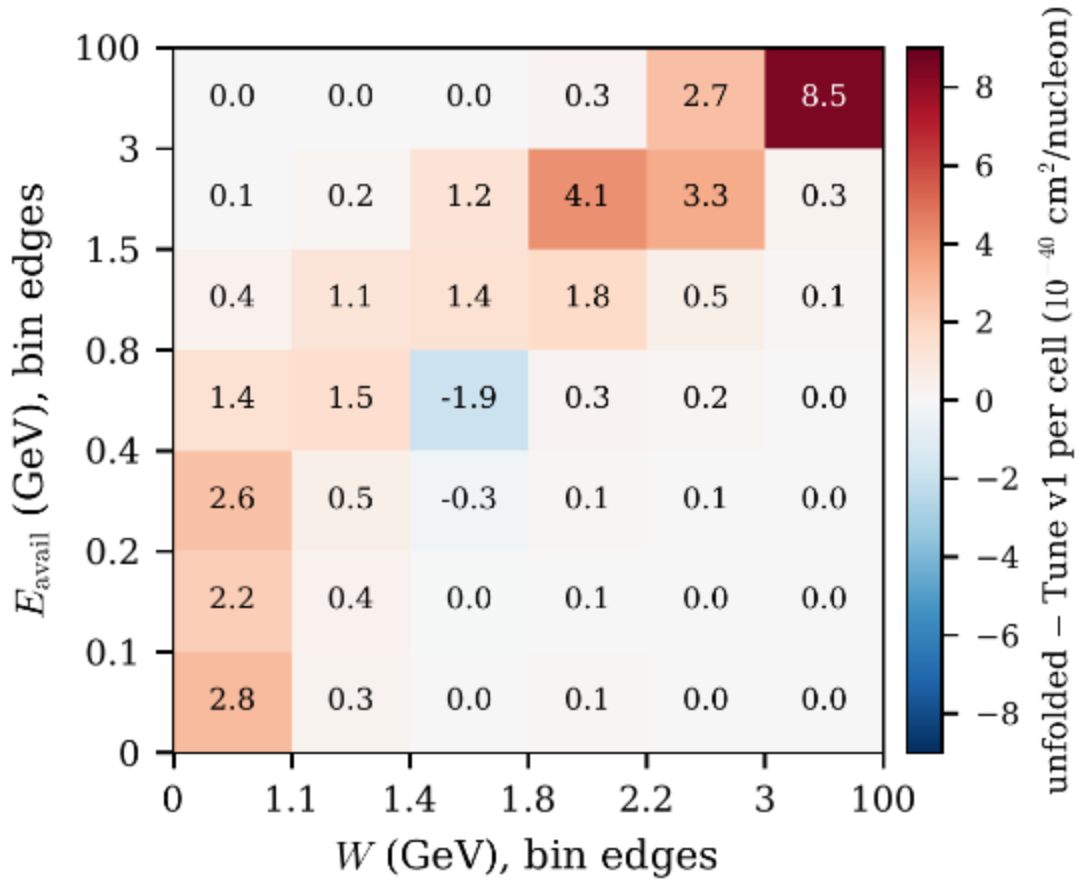
The tested Eavail injection closes

Valencia 2p2h fills 52% of the gap at Eavail ≤ 0.4 GeV

Central-value comparison only

Generator predictions use repaired samples. Historical covariance significances stay unquoted

The excess concentrates at high Eavail and high W



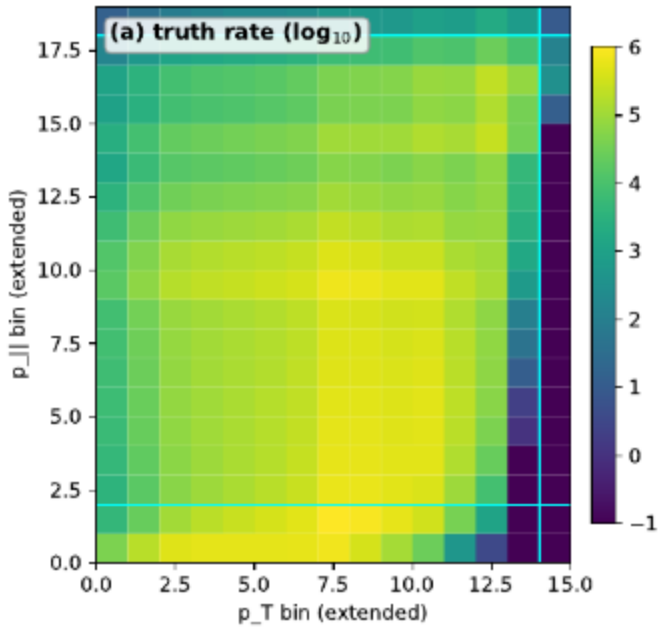
67% of positive excess at Eavail ≥ 0.8 GeV

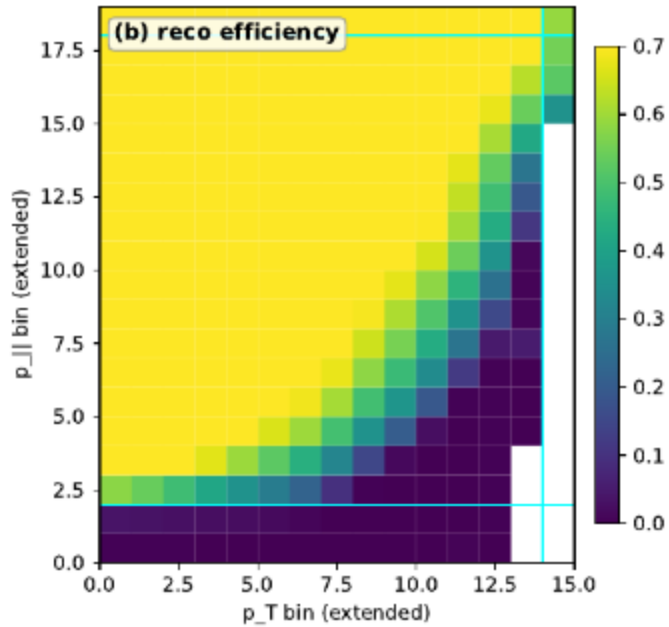
83% of that at W ≥ 1.8 GeV

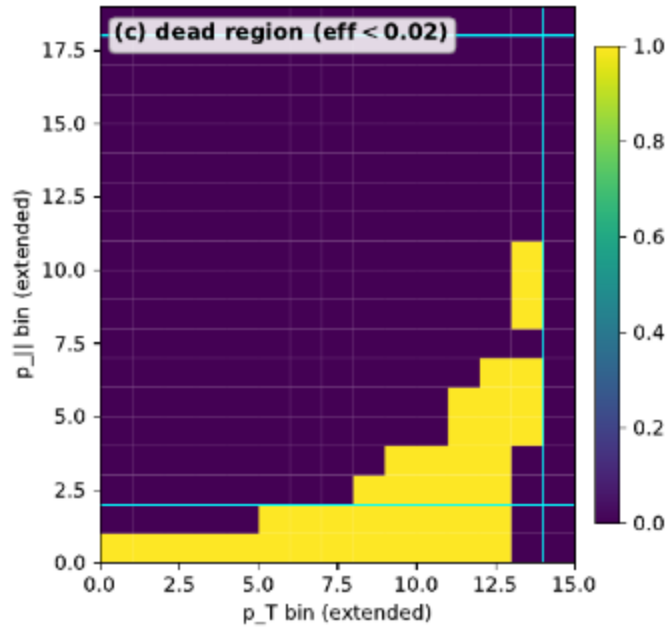
Largest cell: 22% of this map's positive excess

Cell-integrated central differences from MINERvA Tune v1. No ranking of significance across observables.

Extending truth acceptance increases extrapolation







Remove truth-muon kinematic cuts

27.7% of truth rate lies where efficiency < 2%

Keep the reconstructed MINOS selection

Central-value and prior tests only

Extended fiducial domain. No corrected covariance for this extension

The 5% bound limits seed-induced movement of error bars

For independent Gaussian throws

$$SE(s) / \sigma \approx 1 / \sqrt{[2(N - 1)]}$$

N = 160: about 5.61%

Chosen tolerance δ = 5%

Seed-invariant ideal: 0%

Width for a projection u

$$\sigma(u) = \sqrt{(u^T C u)}$$

$$\max |\sigma new(u) / \sigma ref(u) - 1|$$

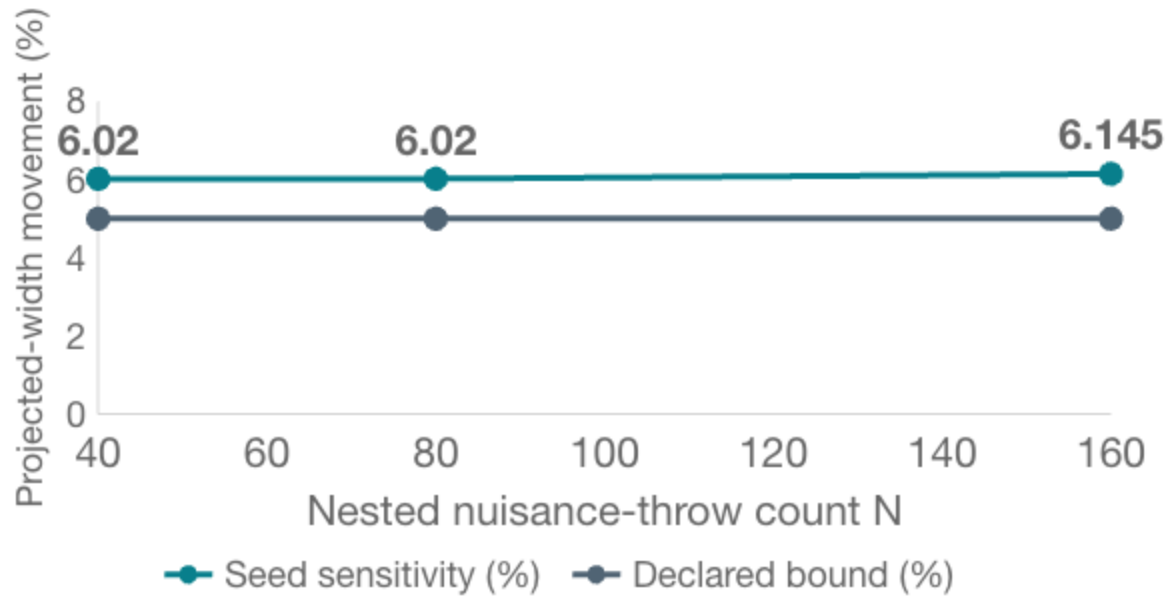
Measured: 6.145% > 5%

6.145% is movement of an error-bar width when the estimator seed changes, with nuisance throws held fixed.

5% is a declared stability tolerance, motivated by sampling precision. It is not a p-value or a simultaneous confidence bound.

Four measurements accompany the adopted 5D covariance

M1 / M2: exceeds 5% throughout this tested range



M3: five seed-pinned bands

6.75% of covariance trace. Their seed effect is unprobed in either direction.

M4: central values also move

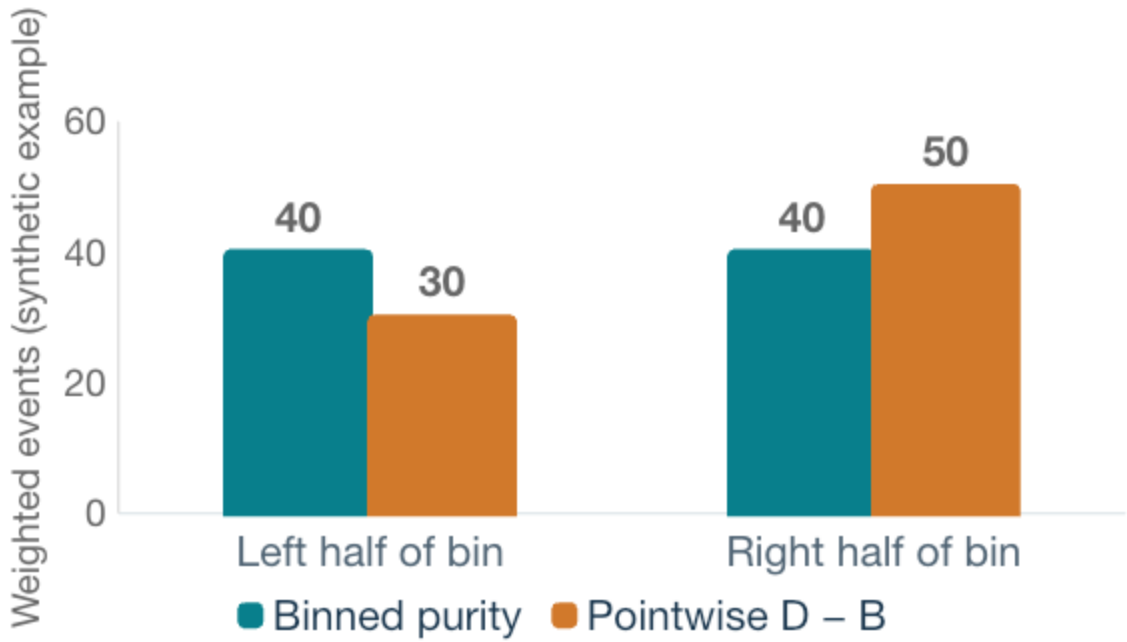
Up to 6.02% of projected σ.
Up to 49.8% of individual-bin σ.

At one seed pair, width movement stays near 6%. The same-seed resampling floor falls from 20.91% at N = 40 to 7.57% at N = 80.

Nested subsets of 160, one seed pair. Larger N unmeasured. Adopted 3d7465f6… remains NON-PASSING; its production C3 was not computed.

Purity weights and negative-weight subtraction

Example: D = (50, 50), B = (20, 0)



Standard 2D / scalar 5D chain

wdata = max(0, 1 – Bb/Db)
Same weight within each reco bin.

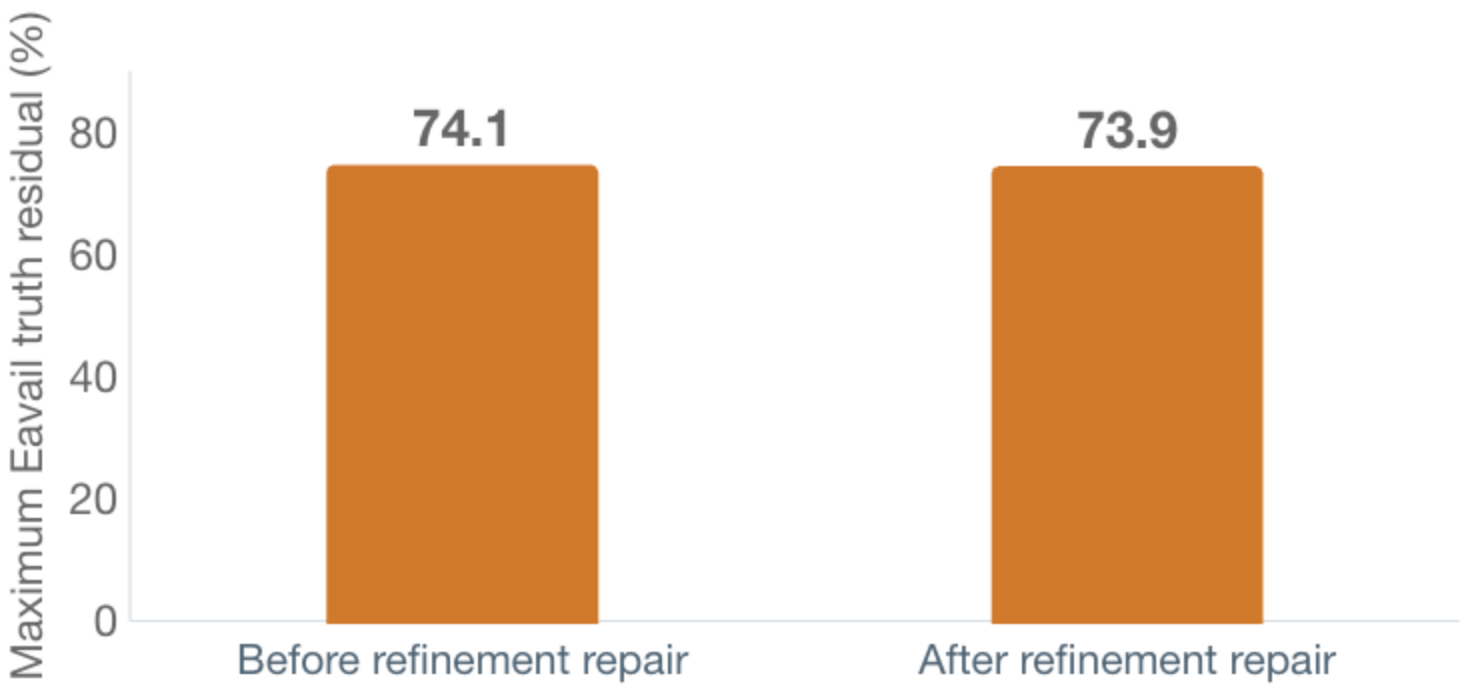
Comparison and later diagnostics

Data weight +1, background weight –wB. Positive refinement precedes the later candidate's fit.

Both totals are 80. Their within-bin shapes differ unless B/D is constant.

Illustration only. Raw signed training, positive refinement, and clipping negative GENIE systematic weights are distinct operations.

Repairing background bias did not repair truth recovery



Background bias ↓

More than 80% reduction in every worst cell.

The truth-departure residual barely changes.

Refinement capacity repairs nominal background bias. It does not resolve the truth-recovery failure.

s5e candidate R, tested Eavail departure. Residuals persist with signal only. Candidate also failed real-data numerical stability; no adoption.

A separate test asks whether specified models fit the data

Specify a hybrid null

Generator fine-cell rates
+ within-cell shape
+ detector and nuisance model

Repeat experiments

Apply the same estimator and calculate the chosen statistic.

Compare with the data

Calibrate a Monte Carlo p-value. Correct the family of ten tests.

Unbinned event unfolding

Finite-grid hypothesis and test

The joint-5D measurement did not qualify. A separately calibrated model test answers a narrower question.

Binning makes the hypothesis and calibration explicit. An unbinned test would require its own statistic and calibration.

All ten specified joint-5D null hypotheses are rejected

Prediction	Total	Shape	Conditions carried by each rejection
GENIE CV	Reject	Reject	Fine-grid rates, imposed within-cell shape
MINERvA Tune v1	Reject	Reject	Declared detector bands; no recoil scale
GENIE + Valencia MEC	Reject	Reject	34 interaction bands; two clipped weights
NuWro	Reject	Reject	Half-MC toys, full-MC fit; shift included
GiBUU	Reject	Reject	GiBUU Ev ≤ 20 GeV; all ≤ 100 GeV
			Generator MC statistics in metric and draws

Holm familywise α = 0.05. Rejection concerns the full specified null, not a uniquely identified generator defect.

s5p conditional inference with candidate R. No new measured cross section or uncertainty. Independent recomputation agrees on the calculation.

The rejection survives a bounded refinement stress test

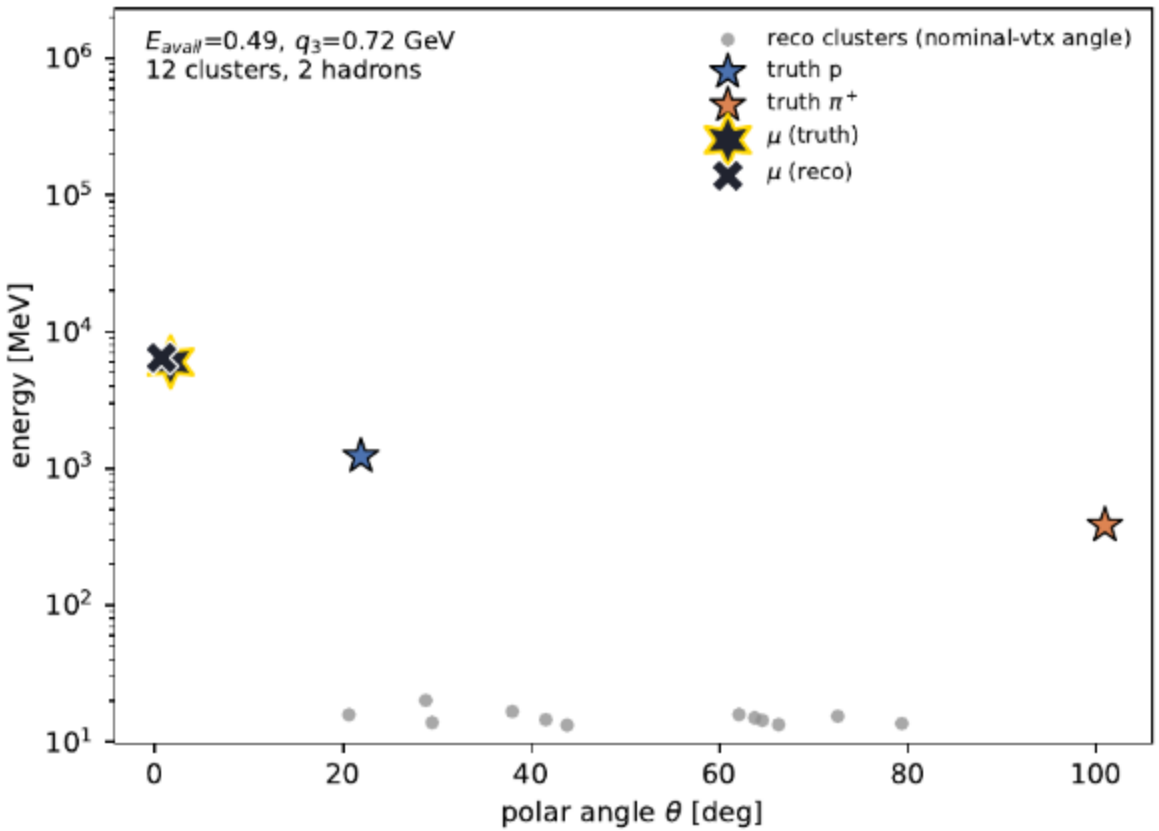
κ multiplies the last change seen when splitting truth cells more finely.



The matched coarse muon-only test does not reject GENIE CV. The added joint coordinates reveal a disagreement.

Refinement has not converged. κ is a stress scale, not a confidence level. Recovered draws shift the first-loss value to ≈5.17. Muon check is post hoc.

PET uses particle and detector-object structure



Objects carry structure

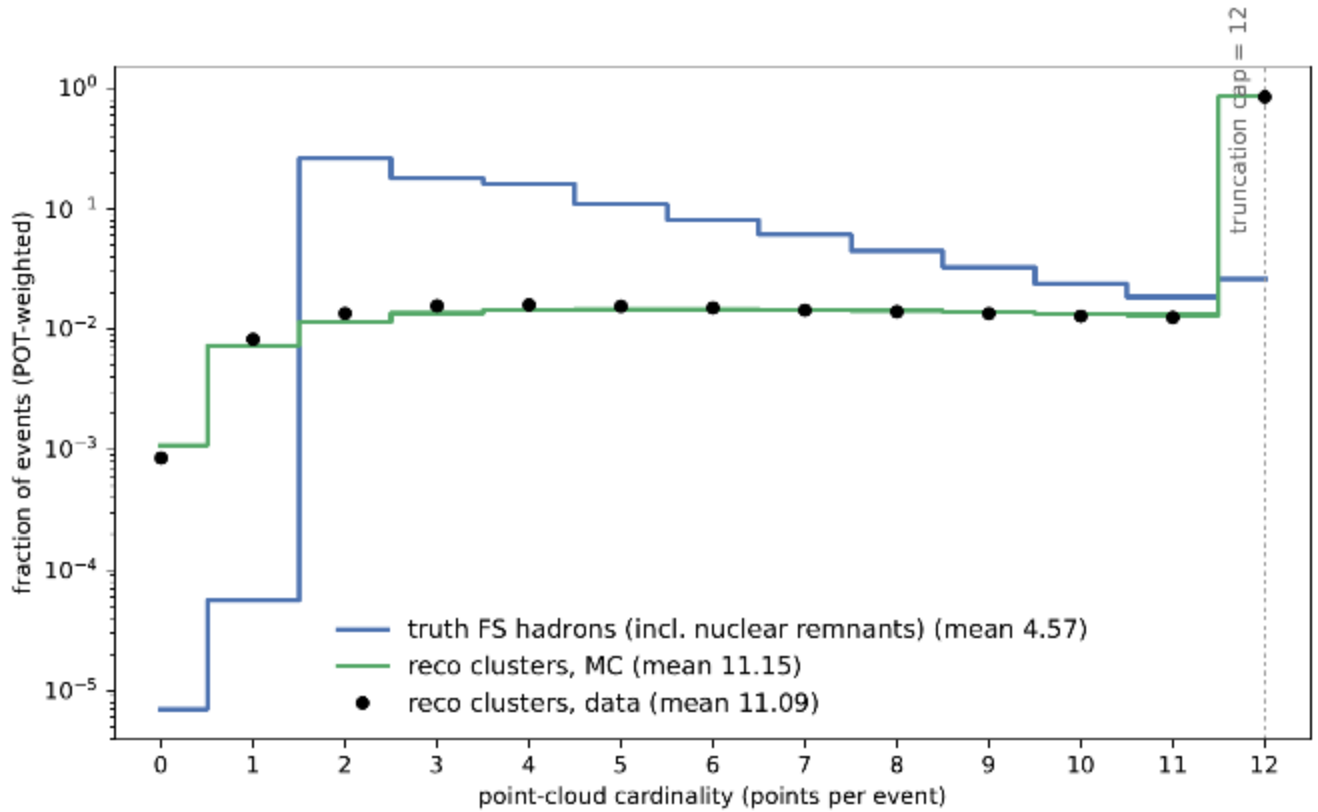
Many detector clusters can come from one hadron.

Muon information remains available separately.

PET learns relationships among objects. The detector cloud is not a list of reconstructed truth particles.

Archived recoil-cloud example. Cluster angle uses a nominal vertex for display; PET inputs use stored view/position features.

The 12-token cap truncates reco clouds much more often



Reco hits the cap

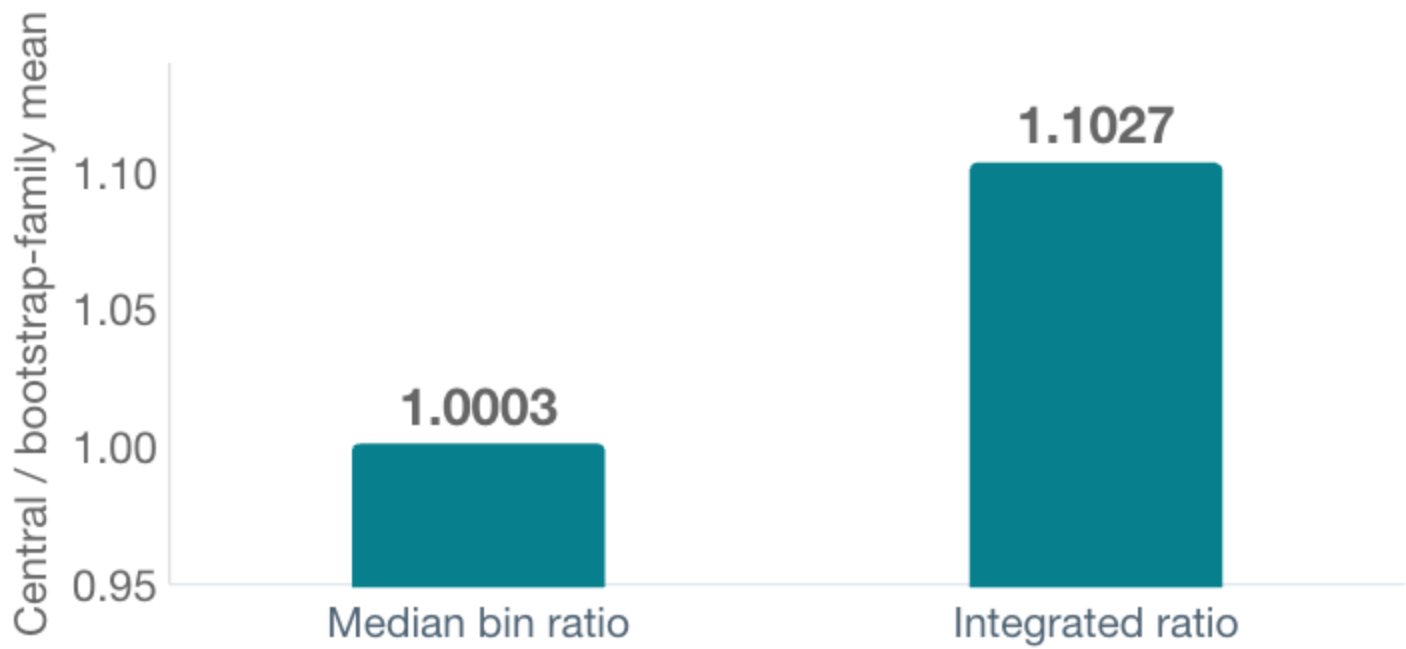
Truth: 2.31% reach 12.

The 12th truth object carries 0.09% of retained energy on average.

Twelve limits compute and memory. A controlled cap scan has not established that it is optimal.

0.09% is not the total discarded energy or lost-information fraction. This post-truncation plot cannot show the uncapped reco tail.

The earlier PET central disagrees with its bootstrap family



Localized mismatch

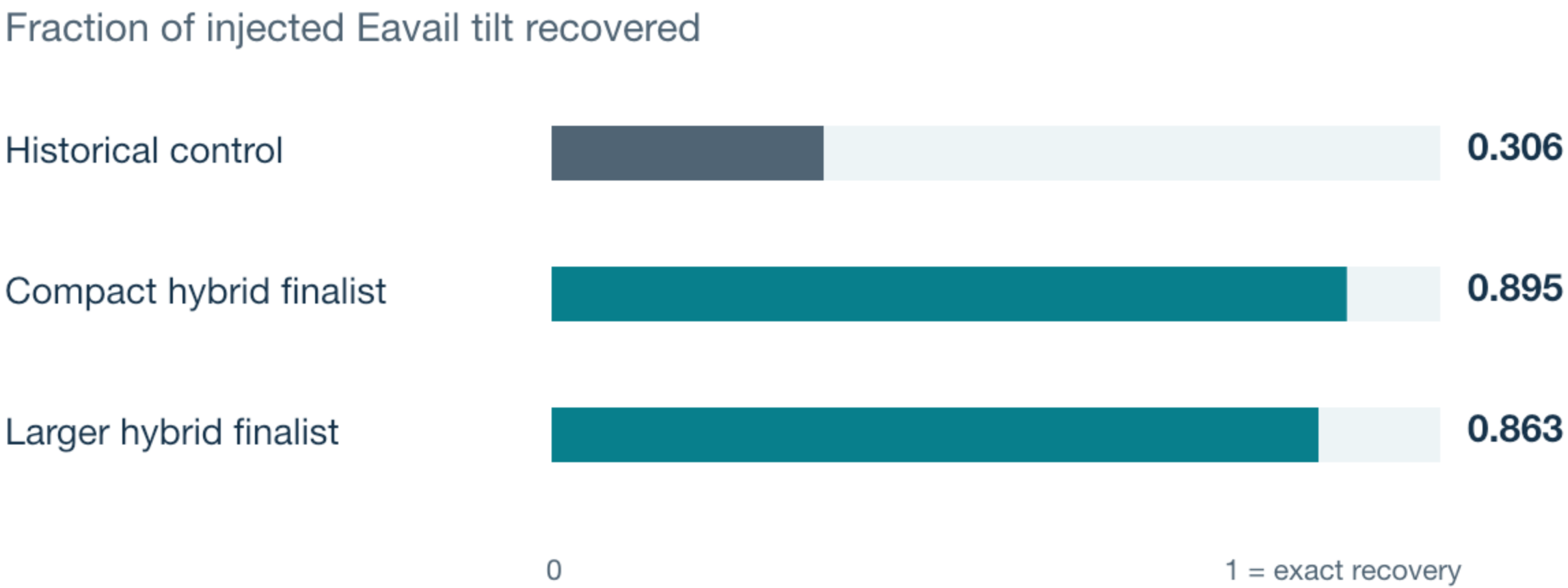
63 of 257 cells have central / mean > 1.5.

In 44 of those, central is above every one of 50 replicas.

The central/statistical pairing was declined. The mechanism remains unresolved.

Corrected same-domain, bin-width-weighted comparison. Covariance exists but is unverified. No adopted PET total covariance or ML block.

PET recovers more of the tested Eavail tilt

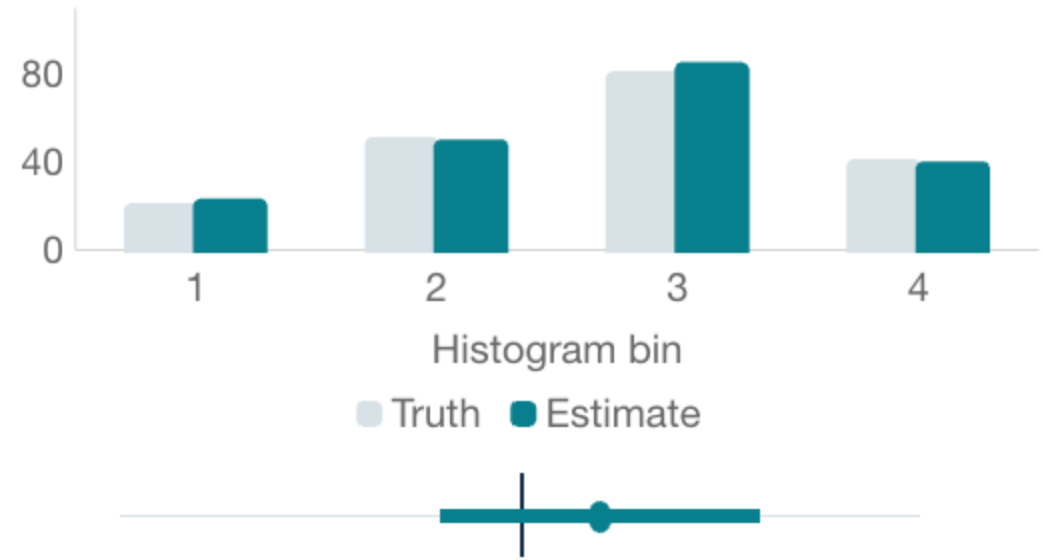


Final-bank Eavail tilt, 60 draws. R = 1 is exact recovery, R = 0 retains the prior discrepancy

Coverage counts intervals that contain the true bin value

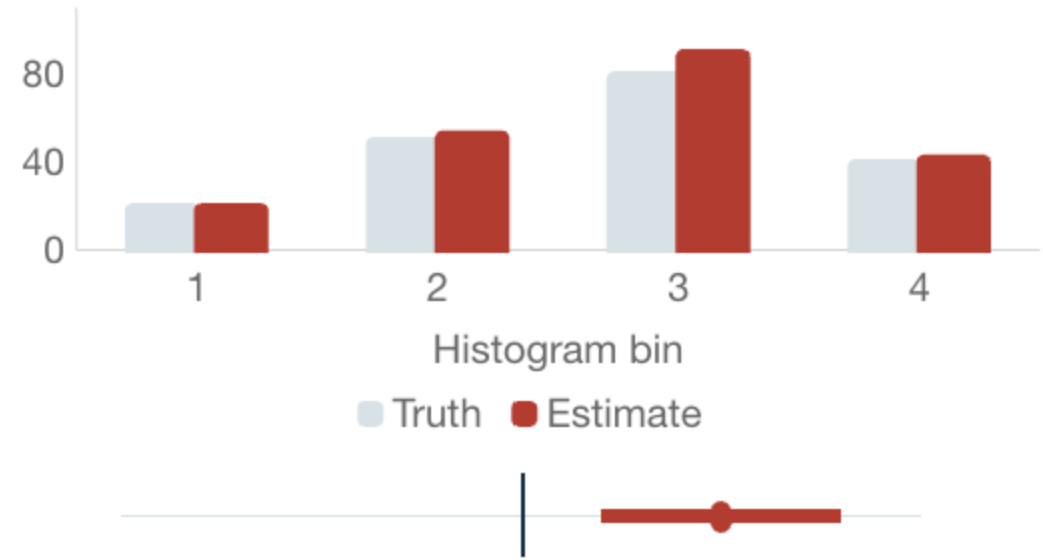
Known truth in bin 3: 80

Experiment A



[76, 92] contains truth = 80

Experiment B

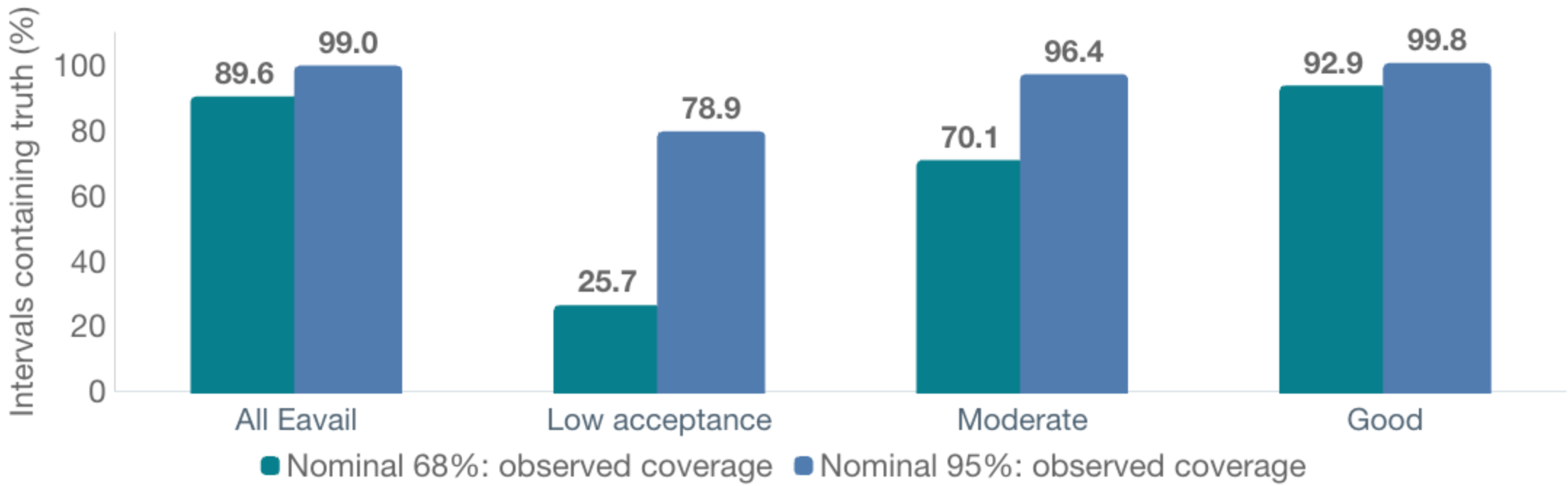


[84, 96] misses truth = 80

Repeat experiments and count containing intervals. Nominal 68% targets 68% containment.

Synthetic illustration, not MINERvA data or a coverage measurement. The intervals apply to bin values, not event labels.

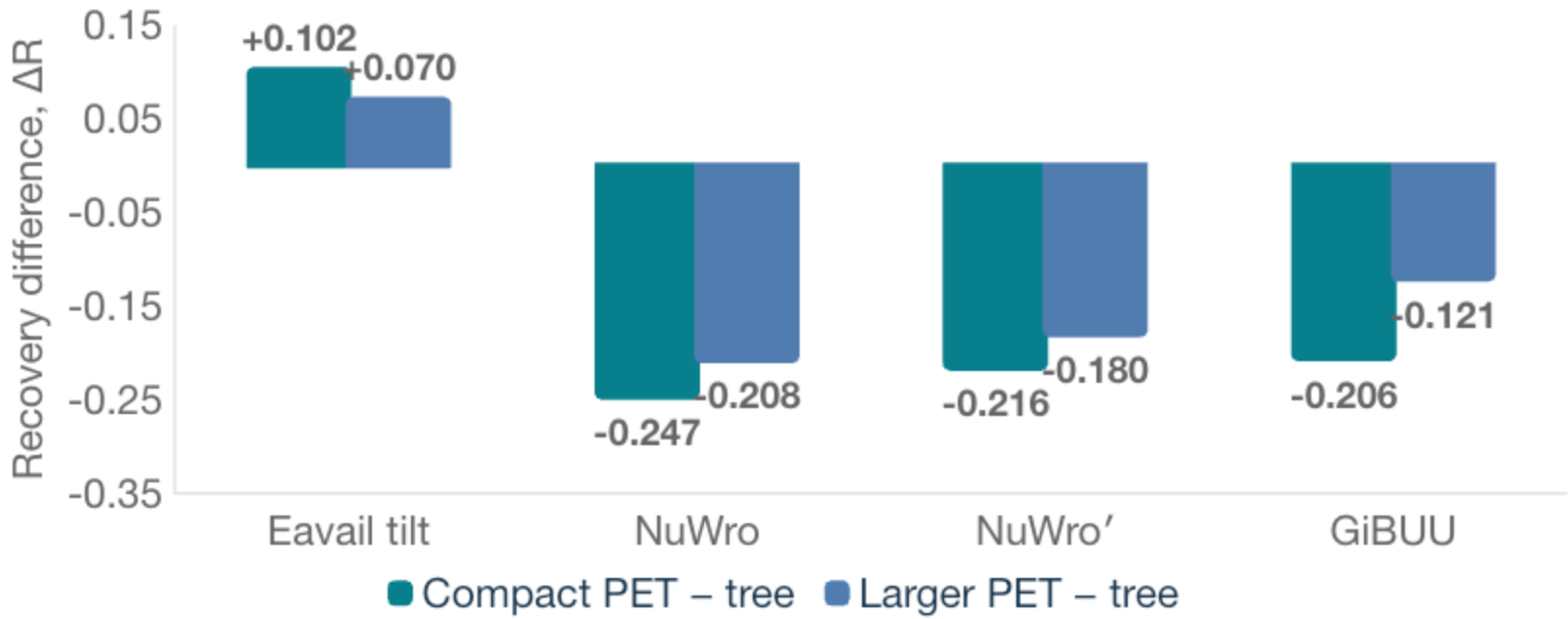
PET intervals over-cover overall and under-cover low acceptance



Low acceptance: 25.7% containment where 68% is intended. Aggregate over-coverage hides the local deficit.

120 pseudo-experiments, six bootstrap members per estimator. Entries pool bin-draw checks, not simultaneous coverage or classification rates.

PET gains depend on the tested departure



Gain depends on the departure

Above zero: PET recovers more.

Below zero: the tree comparator recovers more.

Measured PET cost: 1.768 / 1.517 A100-hours per unfolding at study scale

Means: 60 tilt draws, 8 per generator. Trees: 352 tasks + control/retries, 5.306 charged CPU-core-hours total. Different hardware and packing.

Reproducibility and scientific validation answer different questions

Provenance	Did the intended code and inputs actually run? Hashes alone once missed a wrong-checkout import.
Numerical reproduction	Can a fresh calculation recover the stated result? XR now does this for the LightGBM reference centrals.
Scientific validation	Does the estimator recover truth, and do its intervals cover? Numerical agreement cannot establish either.

Each result needs the check that answers its scientific question. A generic PASS cannot replace that evidence.

The next decision is what measurement claim can be supported

Resolved by XR so far	LightGBM seed-42 and seed-1 reference centrals regenerate within tolerance.
Still unresolved	Exact-tree comparisons in the interim snapshot; covariance transfer; coverage of the corrected 2D band.
Scientific choice	Adopt a matched estimator with explicitly narrower claims, or obtain the independent inputs needed for stronger validation?

The article is assembled, not submitted. The proposed matched-estimator route is not an adopted measurement.

XR 9be8bfb4, 11 October 11:20 UTC. Later evidence should update this snapshot before the talk.

Backup: exact available-energy convention

Truth	Protons: E – 938.27 MeV. Charged pions: E – 135 MeV
Truth continued	Neutral pions and photons: total energy. All other species contribute zero
Reconstruction	1.17 × (tracker + ECAL blob energy after muon-fuzz subtraction)
Comparison limit	The inherited charged-pion constant and species exclusions differ from some published definitions

Backup: the adopted covariance's exact boundary

Identity	z-cv.npz, SHA-256 prefix 3d7465f6, 890,500,272 bytes
Four measurements	M1–M4 on slide 16 travel with every use of this product
Unresolved criterion	Cause 3 remains predeclared and not computed for this digest
Decision meaning	Exception covers these bytes only. No inherited adoption for other candidates

Backup: reading the conditional p-values

Finite simulation	p = (k + 1) / (B + 1), with k exceedances in B toys
Resolution	B = 1,343–1,751. Never report p = 0 or turn it into a Gaussian significance
Test domain	109 supported joint cells. 72 for GiBUU with $p_{\parallel} < 6$ GeV/c
Missing draws	224 calibration + 53 power draws recovered. Frozen decisions unchanged

Backup: the PET representation studies

Pooling versus individual objects	Synthetic routing comparison found no resolved accuracy difference
Production cloud	The top-12 cap removes substantial reconstructed activity
Improvement campaign	No frozen ten-iteration candidate passed adequacy plus stress
Final design and pretraining	No tested PET2 recipe passed every development screen

Backup: the closed scalar sequence

s5c	Futility stop. Nominal purity-background bias confirmed
s5n	Refined subtraction improves nominal behavior. Departure failure persists
s5e	Diagnosed refinement underfitting. Candidate R fails fresh assessment
s5p	Measurement not admitted. Conditional inference completes with no new uncertainty

Backup: what the release can reproduce

Available	Article sources, analysis note, primer and saved inference inputs
Replay	Recompute conditional inference from sufficient products
Different task	Regenerate training and calibration pseudo-experiments end to end
Release limits	Several closure, uncertainty and sensitivity results remain outside RC4 replay

More events: distinguish three different proposals

Same measured exposure	Current 2D uses all 12 ME-FHC playlists and about 4.12M selected events
Larger training sample	PET used subsamples; more MC can improve support and sampling precision
Different data / selection	RHC or new channels need their own flux, response and signal definition
Current uncertainty issue	More resampling does not create independent validation populations

Current MINERvA release lists ME FHC and RHC; low-energy availability should be checked before planning.